Introduction to Calculations of Heat loss on piping with Solar radiation

Przemyslaw Wiecek

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1 Introduction

At some point during one of the project I was a part of, it was necessary to check if the chosen thickness of insulation for long CIP water loops was enough, to keep the temperature control and over certain threshold. It turned out that one of calculations were done in a rush, resulting in wrong decision of making insulation thicker. Just in time, one of senior engineers stepped in and stopped the company from making somewhat pricy mistake.

This senior engineer, later send us the calculaitons to check it, write down things that were wrong and report back to him, to have a conversation about our findings. This way we could learn to spot errors. Because I like calculations and love learning about things inherently related to mathematics it was a match made in heaven. At this point in time I have not started my introductory book to mechanics of heat transfer by great J. Cengel, so it wasn't something I could jump right into. Thus, I took my time, and in around 2 weeks felt somewhat comfortable, at least with concepts (side note: I was especially taken by transient analysis and eigendecomposition of differential equations in plug flow problems, amazing, it might be something to write about) in steady state analysis to begin deriving equations and testing my knowledge.

The original calculations were made using iterative method, which is fine if the iterations are "fairly" small or the change in the underlying function is relatively insignificant or, even better, constant.

Funny note: I'm writting that I felt comfortable in around 2 weeks, but it took me around 5-6 months to understand that my first solution was completely wrong mathematically. Additionally, I had to rewritte my whole code because I convinced myself that all-functions approach is the way to go, which made my code unusable. Then I had to rewritte it again because of abovementioned mathematical error. All in all it took me around 9 months to have working template with this written as well.

2 Basic model and notation

We will be dealing with a lot of variables, and it is important to explain the notation correctly in the beginning. My main basic model assumption is that going from the inside of a piping we have fluid (f), piping wall (w), insulation layer (i), then surface of insulation (s) and ambient air (a). Because we will be deriving general equations that take into account both external heat flux as well as intergnall heat tracing, we will denote a sliding layer (y) that changes from (i) to (s). It is shown on below diagram.

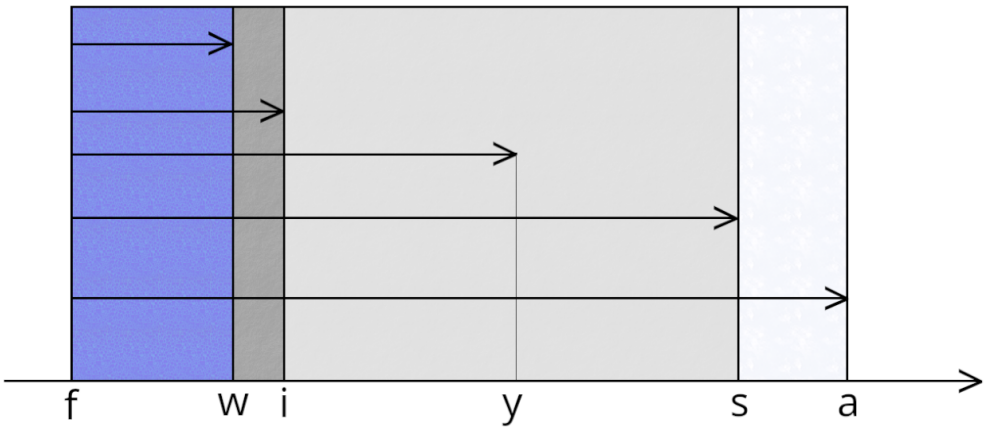


Figure 1: **Piping layering** Shows how layers change with radius. From left: fluid, wall, insulation and air.

Now, let F_{ij} be any variable, then ij index denotes that te variale is measured in relation to layers from i to j. For example, R_{fi} is the combined thermal resistance of fluid and wall layers.

Fluid and surface temperature are a function of length only, even if diameter changes or insulation thickness from one segment to another, those changes are also dependent on length. This is worth keeping in mind, as the dependence will be mostly ommitted.

The value of thermal transfer coeff. around piping surface h_{sa} (h for short) depends on the surface temperature T_s of single piping segment only because of used correlations. This will be always explicilty stated. Of course, as mentioned above surface temperature is a function of length, but in h this will be ommitted for simplicity, and I just think it is easier to think of it as being a function of surface temperature only and not add length dependency.

3 Derivation

3.1 Basics of steady-state heat transfer

The assumptions we must consider to make calculations possible are:

1. Steady-state process.
2. The change in kinetic and potential energy is negligible.
3. The flow is turbulent and pipes are full of liquid.
4. The fluid is incompressible.
5. The specific heat capacity is constant.
6. Fluid is loosing energy to surroundings.
7. No axial heat transfer.
8. Uniform heat transfer distribution in piping cross section.

9. Piping is horizontal and enclosed by surface at ambient temperature separated by air only at ambient pressure and temperature.
10. Internal fluid heat transfer coeff. is insignificant and material conductivities are constant.
11. Absorptivity is equal to unit (heat flux mode).
12. Tracing is at the piping-insulaiton interface (heat tracing mode).

With above in mind, one can utilize Thermal Resistance Network methodology and write that heat transfer in each layer of piping is equal to:

$$\dot{Q}_{ij} = \frac{\Delta T_{ij}}{R_{ij}} \tag{1}$$

The place at which we apply heat flux is different in each calculation mode. In heat tracing we apply heat flux at (y=i) and in external heat flux we apply it at (y=s), because of that the general heat balance can be written as:

$$-\frac{\Delta T_{fy}}{R_{fy}} = \dot{m} C dT_f \tag{2}$$

This will be the form from which we will derive the general equation in the next section. It is also important to define thermal resistance. In general, the total thermal resistance is:

$$R_{fa} = R_{fw} + R_{wi} + R_{is} + R_{sa} = \frac{1}{2\pi L}\left(\frac{2}{h_f D_i} + \frac{1}{k_{pipe}}\ln\left(\frac{D_o}{D_i}\right) + \frac{1}{k_{ins}}\ln\left(\frac{D_t}{D_o}\right) + \frac{2}{hD_t}\right) \tag{3}$$

where:

- L - piping of length L, for which thermal resistance is calculated,
- D_i - inner piping diameter,
- D_o - outer piping diameter,
- D_t - total diameter $D_o + 2 \cdot$ ins.thickness,
- h - combined heat transfer coefficient from surface of a piping to air,
- k_{pipe} - thermal conductivity of a piping,
- k_{ins} - thermal conductivity of a insulation,

Here, because of assumption (10) we let $h_f \to \infty$ in return we have $R_{fw} \to 0$. Thus, form now on we will be writting R_{wi} simply as R_{fi}.

3.2 General solution

3.2.1 Energy balance preparation in finite step temperature change

We will start with analysing finite step analysis of fluid temperature drop over finite length change. Using graph for this would be the simplest approach.

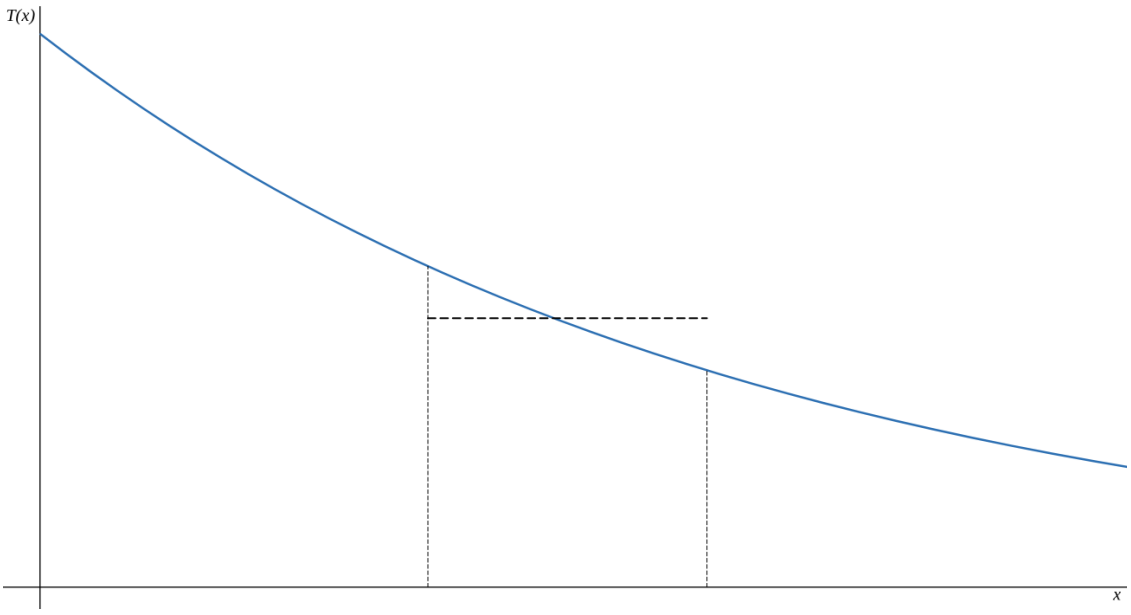


Figure 2: **T(x)** Function of T(x)

In the figure above we consider a change in temperature over a finite change in length on interval $[x_i, x_{i+1}]$, where temperature drops from $T_f(x_i)$ to $T_f(x_{i+1})$. The heat loss over piping segment $x_{i+1} - x_i = \Delta x_i$ is not constant, but drops with temperature. Because of that we consider some average temperature difference to find the average heat loss. The temperature we will use to find the difference is shown on the figure above as a horizontal line between x_i and x_{i+1}. With $w_i \in [0, 1]$ we can write that the average heat loss is:

$$\dot{Q}_{avg}(x_i) = \frac{\Delta T_{fy}}{R_{fy}} = U_{fy} P_t \cdot \Delta T_{fy}(x_i + w_i \Delta x_i) \cdot (x_{i+1} - x_i) = U_{fy} P_t \Delta T_{fy}(x_i^*) \Delta x_i$$

where, for completeness, U_{fy} is the heat transfer coeff. from fluid to sliding layer taken in referecne to total area A_t:

$$U_{fy} = \frac{1}{A_t R_{fy}}, \quad 2\pi L R_{fy} = \frac{1}{k_{pipe}} \ln\left(\frac{D_o}{D_i}\right) + \frac{1}{k_{ins}} \ln\left(\frac{D_y}{D_o}\right)$$

energy balance is:

$$-\dot{Q}_{avg}(x_i) = \dot{m} C \left(T_f(x_{i+1}) - T_f(x_i)\right) = \dot{m} C \Delta T_{f,i}$$

We connect above equations, rearrange it and write:

$$\frac{1}{U_{fy} \Delta T_{fy}(x_i^*)} \frac{\Delta T_{f,i}}{\Delta x_i} = -\frac{P_t}{\dot{m} C} = -\frac{1}{\lambda}$$

We can now take limit at $\Delta x_i \to 0$ and we will arrive at derivative form:

$$\frac{1}{U_{fy} \Delta T_{fy}} \frac{dT_f}{dx} = -\frac{1}{\lambda} \tag{4}$$

And this is basically the equation we must integrate over. Here, the problem is that, we don't know how T_f and T_y are connected to each other and to the applied heat flux. I found that the simplest way is to take cacalculations to the surface temperature domain, as it can be easily connected with all variables. We can then integrate over surface temperature and just translate back to fluid temperature.

3.2.2 Surface temperature domain change and solution

We start by going back to figure (1) and drawing two control systems.

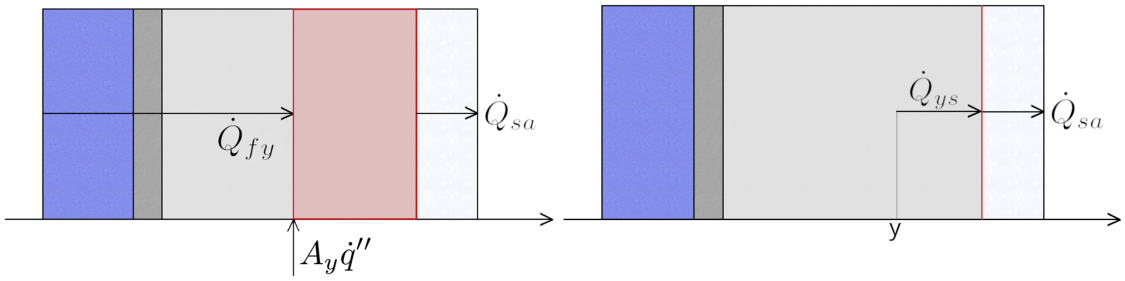


Figure 3: **Control Systems** (1) Control volue at sliding layer; (2) Control boundary at insulation surface

The red lines specify the control system. The first one, a control volume, and the second, a control surface (boundary). With those two, we can write a system of equations fixing the state:

$$\begin{aligned} A_y\dot{q}'' + \dot{Q}_{fy} &= \dot{Q}_{sa} \\ \dot{Q}_{ys} &= \dot{Q}_{sa} \end{aligned}$$

We can substitute eq. (1) for each $\dot{Q}_{ij}$, solve for T_y and substitute bottom equation into the upper one. This allows us to write the general steady state heat transfer equation through piping layers:

$$U_{fy}\Delta T_{fs} + \frac{D_y}{D_t}\dot{q}'' = \left(1 + \frac{U_{fy}}{U_{ys}}\right) h(T_s)\Delta T_{sa} \tag{5}$$

Let us writte the term on the LHS in front of heat transfer coeff. as u_{ry}, this will simplify later calculations. Now, we take differential on both sides getting:

$$U_{fy}\left(dT_f - dT_s\right) = u_{ry}\left(h(T_s)dT_s + \Delta T_{sa}dh_{T_s}\right)$$

and then,

$$dT_f = \left(1 + u_{ry}\frac{h(T_s)}{U_{fy}} + u_{ry}\frac{\Delta T_{sa}}{U_{fy}}\frac{dh(T_s)}{dT_s}\right) dT_s \tag{6}$$

One more thing is to specify $U_{fy}\Delta T_{fy}$, which is required by (4). We can take it simply from the first equation in the system of equations presented below fig. (3). And that's basically it. We substitute it and eq. (6) into eq. (4) and then integrate to get:

$$\int_{T_{s,0}}^{T_s} \left(h(T)\Delta T - \frac{D_y}{D_t}\dot{q}''\right)^{-1} \left(1 + u_{ry}\frac{h(T)}{U_{fy}} + u_{ry}\frac{\Delta T}{U_{fy}}\frac{dh(T)}{dT}\right) dT = -\frac{x}{\lambda} \tag{7}$$

The general heat balance (2) requires fluid temperature, which means that eq. (5) and (7) are nothing more than a set of parametic curves in (T_f, T_s, x) and their intersection satisfies (2). We construct a loss function $\mathcal{L}$:

$$\mathcal{L}(T_s, x) = \left(\int_{T_{s,0}}^{T_s} \mathrm{I}(T)dT + \frac{x}{\lambda}\right)^2$$

And we solve it by minimizing T_s on specially constructed interval Y, which will be discussed in the next subsection.

$$T_s(x) = \underset{T\in \mathrm{Y}}{\operatorname{argmin}}\ \mathcal{L}(x, T) \tag{8}$$

This is then plugged back to (5) to recover fluid temperature.

3.2.3 Minimization interval Y

When solving (7) it becomes clear that there are three possible cases for the value of $\dot{q}''$:

1. $\dot{q}'' = 0$
2. $0 < \dot{q}'' < \frac{D_t}{D_y} h(T_{s,0}) \Delta T_{sa,0}$
3. $\dot{q}'' > \frac{D_t}{D_y} h(T_{s,0}) \Delta T_{sa,0}$

The first one (1) is simple to understand, there is no heat flux input. Two other cases (2) and (3) are simply bounds on critical point of integrand I(T_s). For the first case (1) Y is simply the bound one would naturally choose:

$$\mathrm{Y} = [T_a, T_{s,0}] \tag{9}$$

In case of (2), the surface temperature is still decreasing with increasing length, but the lower bound is no longer ambient temperature due to incident heat flux:

$$\mathrm{Y} = [T, T_{s,0}], \text{ where } T \text{ s.t. } h(T)(T - T_a) = \frac{D_y}{D_t}\dot{q}'' \tag{10}$$

In the last case (3), surface temperature increases with increasing length - fluid is heated by external heat flux. The lower bound is initial surface temperature, but upper bound is now different:

$$\mathrm{Y} = [T_{s,0}, T], \text{ where } T \text{ s.t. } h(T)(T - T_a) = \frac{D_y}{D_t}\dot{q}'' \tag{11}$$

It is important to note, that in case of fluid gaining heat from surroundings (inlet fluid temperature is lower than ambient temperature), there are only two cases to consider, either (1) or (3), bacause $\Delta T_{sa,0}$ will be negative or equal to 0, and Y will be (9) or (11).

4 Heat Transfer coefficient of piping surface

Heat transfer coefficient of air film around piping surface is calculated as a sum of radiation and convection effects. This provides a straightforward way of calculating both effects separately. Radiation effect is calculated with Stefan-Boltzman law, whereas convection effect is calculated with Churchill and Chu correlation for horizontal piping in SI units. That is:

$$\begin{aligned} h(T_s) &= h_{rad}(T_s) + h_{conv}(T_s) \\ h_{rad}(T_s) &= \epsilon \cdot 5.67 \cdot 10^{-8} \cdot (T_s^2 + T_a^2) \cdot (T_s + T_a) \\ h_{conv}(T_s) &= 1389 \cdot \frac{\mu(T_{fl}) \cdot C_p(T_{fl})}{D_t} \left(0.6 + 0.321 \cdot \mathrm{Ra}^{1/6}\right)^2 \end{aligned} \tag{12}$$

where the equation for convective heat transfer coefficient assumes Pr = 0.72 for all range of temperatures. Here, T_{fl}, μ and C_p are film temperature, dynamic viscosity and specific heat capacity of air at insulation-air interface, respectively, and D_t is total diameter in inch of piping segment. All of below equations are defined for SI units only. Users can use both Imperial and SI for calculations, and internally the python module uses SI metric for equations (13), (14) and (15).

Film temperature is just a mean value between surface temperature and ambient temperature:

$$T_{fl} = \frac{T_s + T_a}{2}\,[\mathrm{K}] \tag{13}$$

We calculate dynamic viscosity from Sutherland's law for air:

$$\mu(T) = \frac{1.458 \cdot 10^{-6} \cdot T^{1.5}}{T + 110.4} [\mathrm{Pa} \cdot \mathrm{s}] \tag{14}$$

Polynomial relation for specific heat of air is taken from Cengel, Y. A., Boles, M. A., & Kanoglu, M. (2015). "Thermodynamics: An engineering approach" (8th ed.):

$$C_p(T) = 0.9703 + 6.7898 \cdot 10^{-5} \cdot T + 1.6576 \cdot 10^{-7} \cdot T^2 - 6.7863 \cdot 10^{-11} \cdot T^3 \left[\frac{\text{kJ}}{\text{kg} \cdot \text{K}}\right] \tag{15}$$

Above equations (14) and (15) are valid in a range from around 200K to 1800K. There is no limit on upper or lower temperature in the spreadsheet, but below 273K specific heat of air is assumed to be equal to $C_p(273\text{K})$ and above 1800K to be equal to $C_p(1800\text{K})$.

In equation (12) Ra is the Rayleigh number which for air at low reduced pressure (below 0.1) can be calculated from:

$$\text{Ra} = \frac{g\beta(T_s - T_a)D_t^3}{\nu\alpha} = 880381 \cdot \frac{T_s - T_a}{T_{fl}^3 \cdot \mu(T_{fl})^2} D_t^3 \tag{16}$$

Here, g, β and α is the gravitational acceleration, isobaric thermal expansion coefficient and thermal diffusitivty of air, repspectively. The exact equation above is obtained by assuming ambient pressure of 1 atm.